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Report

Forward and Backward Arm Cycling Are Regulated by Equivalent Neural Mechanisms

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Zehr, E. Paul and Sandra R. Hundza. Forward and backward arm cycling are regulated by equivalent neural mechanisms. *J Neurophysiol* 93: 633–640, 2005. First published August 18, 2004; doi:10.1152/jn.00525.2004. It was shown some time ago that cutaneous reflexes were phase-reversed when comparing forward and backward treadmill walking. Activity of central-pattern-generating networks (CPG) regulating neural activity for locomotion was suggested as a mechanism involved in this "program reversal." We have been investigating the neural control of arm movements and the role for CPG mechanisms in regulating rhythmic arm cycling. The purpose of this study was to evaluate the pattern of muscle activity and reflex modulation when comparing forward and backward arm cycling. During rhythmic arm cycling (forward and backward), cutaneous reflexes were evoked with trains (5 × 1.0 ms pulses at 300 Hz) of electrical stimulation delivered to the superficial radial (SR) nerve at the wrist. Electromyographic (EMG) recordings were made bilaterally from muscles acting at the shoulder, elbow, and wrist. Analysis was conducted on specific sections of the movement cycle after phase-averaging contingent on the timing of stimulation in the movement cycle. EMG patterns for rhythmic arm cycling are similar during both forward and backward motion. Cutaneous reflex amplitudes were similarly modulated at both early and middle latency irrespective of arm cycling direction. That is, at similar phases in the movement cycle, responses of corresponding sign and amplitude were seen regardless of movement direction. The results are generally parallel to the observations seen in leg muscles after stimulation of cutaneous nerves in the foot during forward and backward walking and provide further evidence for CPG activity contributing to neural activation and reflex modulation during rhythmic arm movement.

INTRODUCTION

Experiments in various lower animal preparations (e.g., cat and crayfish) have indicated that the same neural mechanisms [e.g., locomotor central pattern generator (CPG)] controlling forward locomotion may be reversed during backward locomotion (Grillner 1981; Pearson 1993). In the cat, research on the pattern of backward and forward walking support the concept that spinal CPGs may simply run in reverse when the movement direction changes (Buford and Smith 1990, 1993; Perell et al. 1993). During human locomotion, it has been shown that for some joints (especially the hip) kinematics and to a lesser extent EMG of backward walking are similar to those of forward walking (Grasso et al. 1998; Thorstensson 1986; Winter et al. 1989). However, electromyographic (EMG) amplitudes are typically higher during backward walking compared with forward walking (Duysens et al. 1996; Grasso et al. 1998; Thorstensson 1986; Winter et al. 1989). The similarities between forward and backward locomotion could reflect the activity of CPG networks running in reverse to produce backward walking. Indeed Grasso et al. (1998) suggest that their data support the activity of similar CNS mechanisms operating to regulate forward and backward walking. It has also been suggested that the same CPG mechanisms may regulate various patterns of gait in the infant (Lamb and Yang 2000).

Modulation of motor activity due to changes in peripheral feedback during rhythmic movement can be used to infer the activity of CPG circuits (Burke 1999; Burke et al. 2001; Zehr and Duysens 2004). For example, the modulation of cutaneous reflexes during rhythmic movement has been suggested to arise due to activity of a human locomotor CPG (Duysens and Tax 1994; Zehr et al. 2001), and this could explain phase- and task-dependency of reflexes via premotoneuronal gating of afferent feedback (Dietz 2002a,b; Duysens and Tax 1994; Duysens and Van de Crommert 1998; MacKay-Lyons 2002). It has been shown that cutaneous and H reflexes are phase and task dependent during arm cycling (Zehr and Chua 2000; Zehr and Kido 2001; Zehr et al. 2003). Further, reflexes evoked by stimulation of cutaneous nerves in the hand and foot are both phase and task dependently modulated in arm muscles during the natural arm swing of walking (Haridas and Zehr 2003; Zehr and Haridas 2003). These observations support the suggestion that rhythmic arm movements are to some extent regulated by CPGs just as posited for the leg (Dietz 2002a,b; Dietz et al. 2001; Zehr and Duysens 2004; E. P. Zehr, T. J. Carroll, R. Chua, D. F. Collins, A. Frigon, C. Haridas, S. R. Hundza, and A. Kido, unpublished data).

Duysens and colleagues (1996) studied cutaneous reflex modulation evoked by electrical stimulation of the sural nerve during forward and backward treadmill walking. Modulation of reflex amplitude throughout the step cycles could be generally explained by a CPG running in "reverse" when going backward. However, the pattern was not strictly simply reversed as some muscles showed minor timing differences and shifts during backward locomotion. This observation was similar to the pattern of cutaneous reflex modulation seen during forward and backward walking in the cat (Buford and Smith 1993). There it was suggested that the central control of cutaneous reflex amplitude was similar for forward and backward quadrupedal walking (Buford and Smith 1993). In this paper, we undertook to replicate this kind of experiment during arm cycling where it is relatively easy to perform a simple

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reversed movement during backward versus forward cycling. For example, both backward and forward leg cycling generate very similar phase-reversed patterns of muscle activity (Eisner et al. 1999; Ting et al. 1999). The purpose of the experiment described in this paper was to test the hypothesis that cutaneous reflexes evoked during backward arm cycling would show a similar but reversed pattern of modulation to that seen in forward arm cycling. Support for this hypothesis would further add to the evidence for CPG contributions to rhythmic human arm movement (Zehr and Duysens 2004) such as has already been demonstrated for the cat forelimb (Yamaguchi 2004).

METHODS

Eleven subjects participated in the experiment with informed, written consent and under the sanction of the Human Research Ethics Board at the University of Victoria.

Protocol

The experimental methodology and protocol are similar to that described in previous experiments involving reflex modulation during walking (Zehr and Haridas 2003) and arm cycling (Zehr and Chua 2000; Zehr and Kido 2001; Zehr et al. 2003). Thus only differences in methodology are highlighted here. Participants performed rhythmic arm cycling using a previously described arm ergometer (e.g., Zehr et al. 2003). Arm cycling was performed in a forward direction (in which clockwise movement of the right arm can be observed from the right side) and then in a backward direction in separate trials.

Nerve stimulation

Electrical stimulation was delivered pseudorandomly throughout the movement cycle to the superficial radial nerve (SR) at the wrist of the right hand with trains of 5 × 1-ms pulses at 300 Hz (Zehr and Haridas 2003; Zehr and Kido 2001) applied with a Grass S88 (Grass Instruments, AstroMed) stimulator connected in series with an SIU5 isolator and a CCU1 constant current unit. The SR nerve was stimulated at approximately twice the threshold for radiating paresthesia through bipolar surface electrodes placed just proximal to the radial styloid on the right arm. Appropriate stimulation location was verified by determining that sensation was evoked in the innervation area of the SR nerve (dorsolateral portion of the right hand).

EMG

Bilateral bipolar recordings were made from shoulder muscles anterior (AD) and posterior (PD) deltoid, elbow muscles biceps (BB) and triceps (TB) brachii, and the wrist flexor carpi radialis (FCR) muscles. EMG signals were preamplified and band-pass filtered at 100–300 Hz (P511 Grass Instruments, AstroMed).

Data acquisition and analysis

Data were sampled at 1,000 Hz with a 12-bit A/D converter connected to a computer running custom-written (Dr. Timothy Carroll, University of New South Wales, Australia) LabView (National Instruments, Austin TX) virtual instruments. Post hoc the movement cycle was divided into 12 equidistant bins or phases that represent positions on the clockface and responses to stimuli in each bin were averaged. "Control data" obtained from cycles without nerve stimulation were used to create subtracted traces (~10–20 observations per bin) of reflex EMG (Zehr and Kido 2001).

EMG analysis

Reflexes were examined at early (~50–80 ms) and middle (~80–120 ms) latencies. During analysis for a given subject all subtracted reflex traces for a muscle (i.e., for all 12 bins) were plotted, and the mean and SD of the prestimulus EMG was calculated as an index of subtraction error. Reflex amplitudes at each latency were considered significant and included in the analysis if they exceeded a 2 SD band calculated from this residual. To quantify the reflex amplitudes, a 10-ms window centered on the peak of each response was calculated at early and middle latency. All EMG and reflex amplitudes were normalized to the maximum control background EMG recorded during forward cycling for each muscle.

Statistics

ANOVA was used to determine main effects and statistically significant reflex amplitudes as well as phase dependency during forward and backward cycling (Statistica, Statsoft). This analysis was conducted on datasets averaged across all subjects. Tukey's HSD test was used to post hoc significant main effects (e.g., significant differences between direction). Linear regression analysis using Pearson's correlation (r) was conducted between reflex amplitudes and background control EMG for each muscle at each phase to determine the extent to which variations in reflex size were linearly related to changes in background muscle activation during forward and backward arm cycling. For this regression analysis all data from each subject were used giving 9° of freedom and yielding a critical r of 0.602 at $P < 0.05$. Descriptive statistics included means ± SE. Statistical significance was set at $P \leq 0.05$.

RESULTS

Rhythmic EMG patterns

Background rhythmic EMG amplitudes were significantly (main effect $P < 0.05$) modulated by phase in the movement cycle for all 10 muscles examined. The pattern of EMG activity (i.e., the amplitude of EMG activity relative to phase in the movement cycle) when examining forward and backward cycling was generally similar for most muscles (see Fig. 1). However, EMG during backward cycling was higher (main effect for direction $P < 0.05$) than during forward arm cycling for half of the muscles studied (iAD, iPD, iBB, cAD, and cBB; see Fig. 1). Note that in Fig. 1 EMG amplitudes are plotted such that the arm position is the same for forward and backward cycling and thus the two directions can be directly compared. To specifically contrast EMG amplitude in the two directions, amplitudes are expressed as percentages of peak EMG amplitude during forward arm cycling. Significant differences between forward and backward amplitudes at each phase (deduced from post hoc for significant interaction, $P \leq 0.05$) are as indicated (*).

Reflex latencies

Time to the peak response for the early and middle latency reflexes are shown in Table 1 for forward and backward arm cycling and did not significantly differ between the movement directions.

Reflex modulation patterns across the movement cycle

Reflex amplitudes at early and middle latency were similarly modulated during forward and backward arm cycling.

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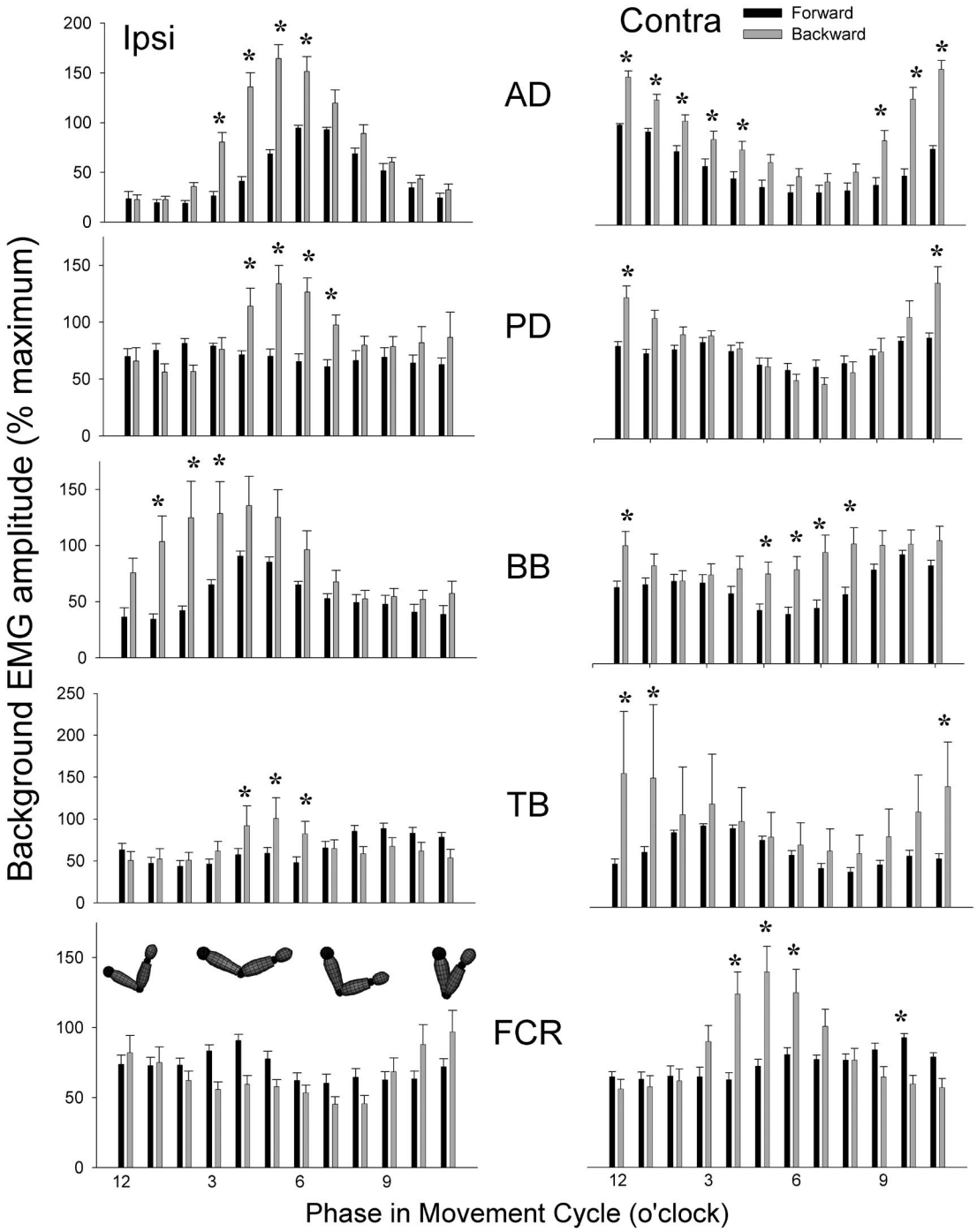

FIG. 1. Pattern of rhythmic electromyography (EMG) across the full movement cycle during forward and backward arm cycling. Maximum background EMGs for all 10 muscles are expressed relative to the maximums during forward arm cycling. Muscles ipsilateral (ipsi) and contralateral (contra) to the site of nerve stimulation are shown on the *left* and *right,* respectively. Phase in the movement cycle relative to the clock face are as indicated at the bottom of the figures. *Bottom left*: the arm position is roughly indicated by the cartoon arm. Data represent means ± SE for 11 subjects. Data are plotted on identical scales for ipsilateral and contralateral muscles. AD and PD, anterior and posterior deltoid; BB, biceps brachi; TB, triceps brachi; FCR, flexor carpi radialis. *, significant differences from post hoc testing for the interaction between movement direction and movement phases at $P \leq 0.05$.

That is, excitatory or inhibitory reflexes were seen at comparable phases in either forward or backward cycling and were of similar amplitude and corresponding sign. Shown in Fig. 2 are reflex traces from iPD muscle for a single subject during forward (*A, left*) and backward (*A, right*) arm cycling. The numbers shown to the *left* of Fig. 2*A* indicate the phase in the movement cycle relative to the clock face. Note that the orientations are the same for forward and backward cycling allowing for direct comparison. The excitatory responses during both directions of cycling are highlighted by the gray rectangles. In Fig. 2*B*, reflex traces during forward and backward cycling are shown superimposed. These traces are from phases (3–4 o'clock) where background EMG levels were similar for this subject and where the arms

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TABLE 1. *Average time to peak early and middle latency responses during forward and backward arm cycling*

	Early		Middle	
Muscle	FWD	BWD	FWD	BWD
iAD	70.4	68.9	106.7	104.3
iPD	76.1	79.1	112.7	114.1
iBB	64.8	72.2	100.9	108.7
iTB	67.0	68.4	98.5	101.8
iFCR	67.5	68.6	100.2	98.3
cAD	66.8	75.4	97.8	110.9
cPD	61.8	72.5	95.7	109.4
cBB	59.0	67.3	94.8	97.9
cTB	64.1	78.5	101.0	118.3
cFCR	61.5	73.2	95.0	105.6

(Data represent mean values for 11 subjects. AD and PD) anterior and posterior deltoid; BB and TB, biceps and triceps (TB) brachi; *FCR,* flexor carpi radialis; *i,* ipsilateral; *c,* contralateral; FWD and BWD, forward and backward arm cycling

where in similar orientations. The similarity between the responses can be seen clearly. It is also observed that the excitatory responses may persist across a greater portion of the movement cycle for backward arm cycling (from ~1 to 6 o'clock) as compared with forward cycling. Note as well that the pattern of responses in PD muscle differs somewhat from the other muscles studied in that a large early latency excitatory burst dominates with only very small responses at middle latency. This can be seen in Fig. 2 for a single subject as well as in Figs. 3 and 4 described in the following text for all subjects.

When the average responses across all subjects were considered, the pattern of reflex modulation was seen to be similar during forward and backward cycling at both early (see Fig. 3) and middle (see Fig. 4) latencies. This was particularly the case for the ipsilateral and more proximal muscles. For example, in ipsilateral AD (iAD), suppressive early latency responses can be seen to occur from 4 to 8 o'clock and for ipsilateral PD (iPD), excitation of varying amplitude predominates across the entire cycle of movement irrespective of movement direction (Fig. 3). As shown in Fig. 4, the middle latency responses switched to excitation in iAD at the same phases for both forward and backward cycling. Further, there was coordination of the responses across the body as can be seen when comparing early (see Fig. 3, *top*) and middle latency (Fig. 4, *top*) responses in iAD and contralateral AD (cAD). This could also be seen when examining ipsilateral and contralateral FCR (iFCR and cFCR) responses at both latencies. The direction of movement had a significant effect ($P < 0.05$) for 4 muscles. Reflex amplitudes in FCR and AD muscles for both arms at

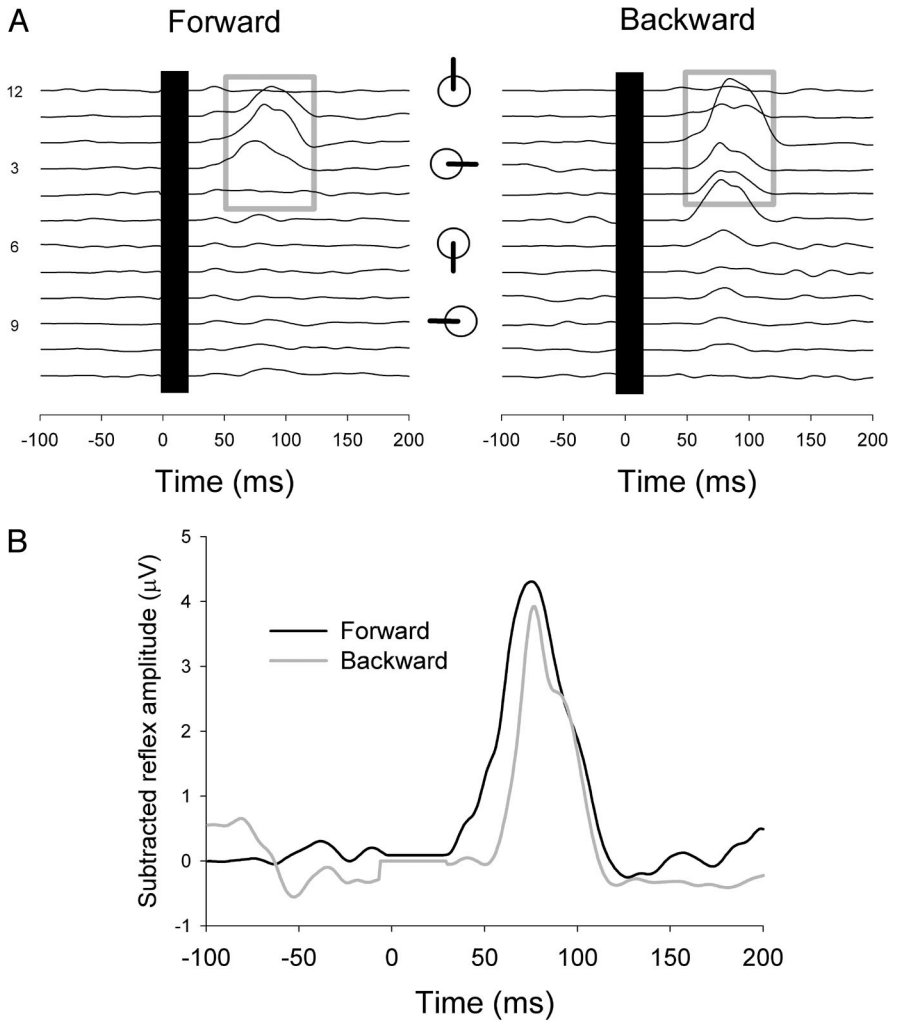

FIG. 2. Cutaneous reflexes in ipsilateral PD (iPD) muscle for a single subject during forward and backward arm cycling. *A*: plots of subtracted reflex EMG for all 12 phases in the movement cycle during forward (*left*) and backward (*right*) cycling. The symbols in between the panels indicate position of the arm cycle hand grip at 12, 3, 6, and 9 o'clock (correspond to numbers given at left of figure). Note the similarity of the facilitation in PD particularly between 12 and 4 o'clock highlighted by the rectangles. This similarity is further emphasized in B in which plots of reflexes from 3 (forward) and 4 (backward) o'clock are superimposed. Note that the stimulus artifact is replaced with a vertical black bar in the 2 panels in *A*.

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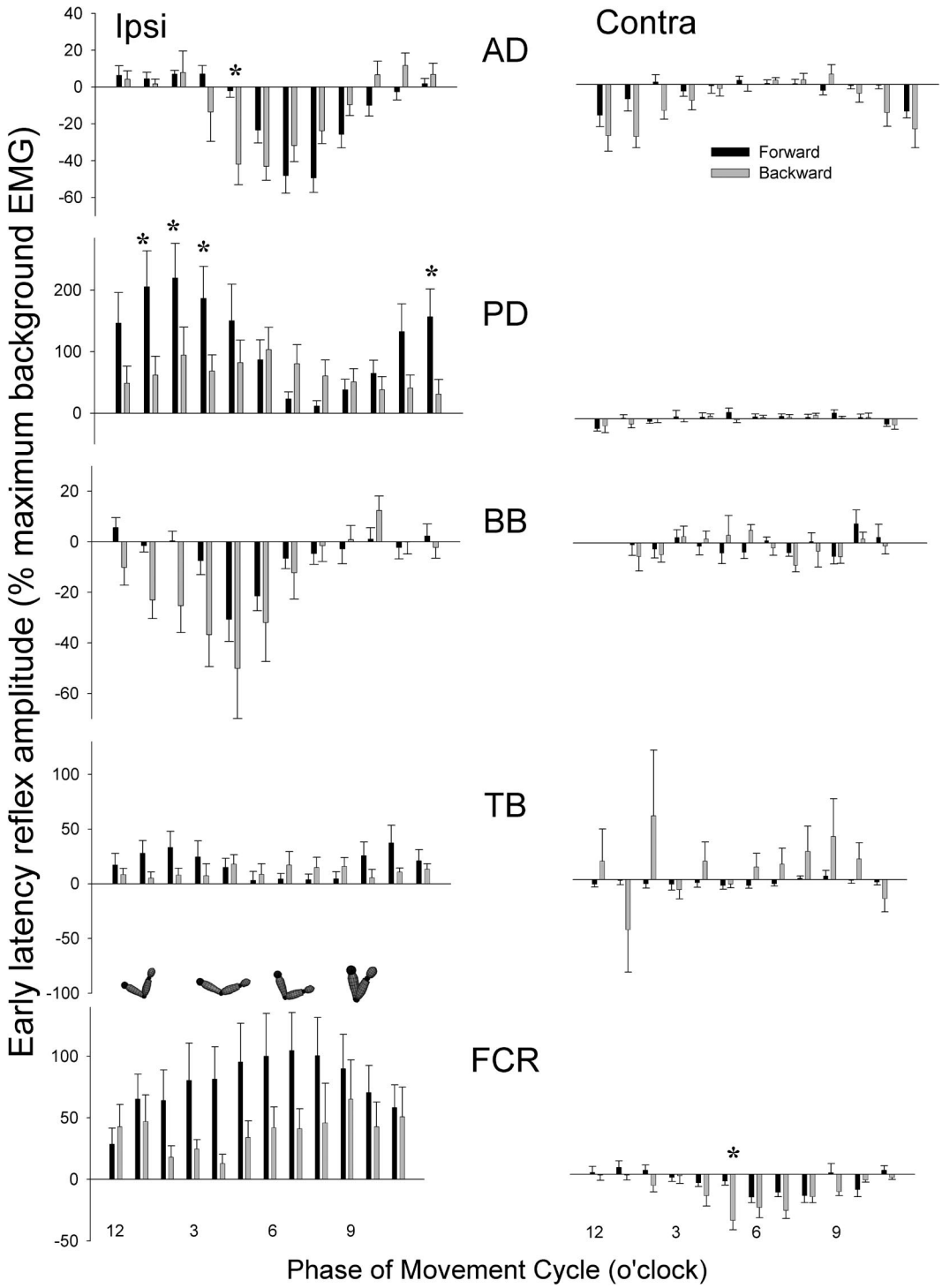

FIG. 3. Early latency (~50–80 ms) reflexes across the entire movement cycle for all subjects. Data are calculated from subtracted reflex traces averaged across all subjects and represent the means ± SE. Data are plotted on identical scales for ipsilateral and contralateral muscles. Abbreviations as in Fig. 1. *, significant differences for the interaction between movement direction and movement phases at $P \leq 0.05$.

early and middle latencies were significantly different when grouped across movement phases (seen as a main effect from the ANOVA; not illustrated on the figures). Even so, the overall pattern of inhibition and excitation was still maintained. At middle latency, there were no main effects for direction of cycling in any muscle examined. Significant interactions between movement phase and direction of movement were identified for early latency reflexes in iAD, iPD, ipsilateral TB (iTB), and cFCR and for middle latency in iAD and cFCR. Results of the post hoc testing ($P \leq 0.05$) for these interactions are shown in Figs. 3 and 4 (*). Note that this analysis identifies if the reflex amplitudes during forward and backward cycling were significantly different at each movement phase. As can be seen there are few in-

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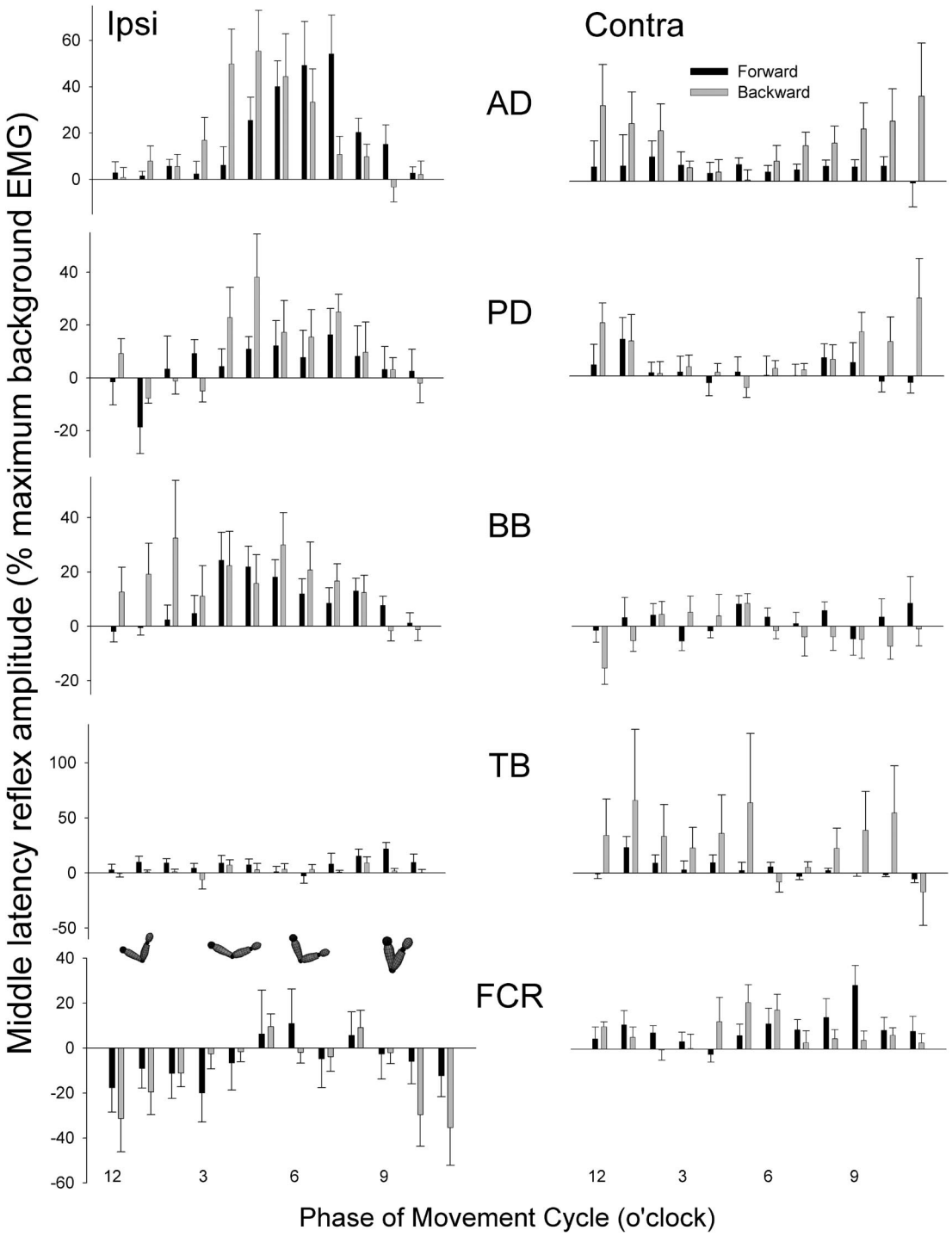

FIG. 4. Middle latency (~80–120 ms) reflexes for all subjects. Data are calculated from subtracted reflex traces averaged across all subjects and represent the means ± SE. Data are plotted on identical scales for ipsilateral and contralateral muscles. Abbreviations as in Fig. 1. *, the interaction between movement direction and movement phases at $P \leq 0.05$.

stances of significant differences between arm cycling direction at any given phase in the movement cycle at the early latency (Fig. 3) and none for middle latencies (Fig. 4).

Previously it has been shown that reflex amplitude is typically uncoupled from rhythmic background EMG amplitude during forward arm cycling (Zehr and Kido 2001). To examine the extent to which reflex modulation during backward cycling was coupled to background EMG during backward arm cycling, we examined Pearson correlations here. Across all muscles for forward and backward cycling and at early and middle latency this analysis yielded 40 comparisons. In only 1 instance was a significant relation with background EMG demonstrated (iAD at early latency during forward cycling). Thus during both forward and backward cycling, reflex amplitude was typically uncoupled from rhythmic background EMG.

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DISCUSSION

There are three main new observations in this paper. First, based on EMG patterns, backward arm cycling represents a simple reversal from forward arm cycling. Second, the pattern of cutaneous reflex modulation during forward arm cycling is similar during backward arm cycling when expressed at similar phases in the movement cycle. Third, the patterns of cutaneous reflex modulation are independent of rhythmic background EMG amplitude during both forward and backward arm cycling.

EMG of forward and backward arm cycling

With some subtle differences the general EMG pattern associated with forward arm cycling was maintained during backward motion. For many muscles studied, the magnitude of rhythmic EMG amplitude during backward arm cycling was significantly greater than corresponding values for forward cycling (e.g., see Fig. 1, *). This is similar to the observations reported for EMG amplitudes during forward versus backward walking (Grasso et al. 1998; Winter et al. 1989). There were some shifts in timing of peak activities of certain muscles and minor shifts in the temporal patterns within the movement cycle. This may be similar to previous documentation of EMG during leg cycling that may be due to subtle changes in the biomechanical function of the muscles during backward motion (Eisner et al. 1999; Ting et al. 1999). However, this was not explicitly evaluated in this study and cannot be definitively discussed here. In general, the pattern of EMG during backward arm cycling was similar to forward cycling (see Fig. 1).

Reflex modulation during backward arm cycling

The pattern of cutaneous reflex modulation during backward arm cycling was equivalent to that seen during forward cycling (e.g., see Figs. 2–4). Across both cycling conditions responses in contralateral muscles were typically of smaller amplitude (as seen in Figs. 3 and 4 in which responses are plotted on the same scale across the body), except for iTB and cTB. There are numerous instances of reciprocal coordination at similar latencies in contralateral muscles and in antagonist muscles (e.g., contrast AD and PD responses in Figs. 3 and 4) just as shown previously for SR nerve reflexes during arm cycling (Zehr and Haridas 2003; Zehr and Kido 2001) and walking (Zehr and Haridas 2003). There were some differences in terms of the timing in the movement cycle of maximum reflex amplitudes and the amplitudes themselves. However, the general features were very similar in that there appeared to be no drastic re-organization of the reflex control such that an entirely new pattern arose with backward cycling as compared with forward motion. That is, reflex responses were still phase-modulated and excitation or inhibition still occurred at similar phases and were not replaced by responses of opposite sign in forward versus backward arm cycling. CPG networks "running in reverse" during backward motion could explain this observation in much the same way that this has been used to explain the observed reversal in the pattern of cutaneous reflexes during backward quadrupedal (Buford and Smith 1993) and bipedal walking (Duysens et al. 1996). Buford and Smith (1993) suggested that, although they did observe some differences in the pattern of cutaneous reflex modulation during backward walking, these differences were very subtle and expressed as minor differences in reflex amplitude and general timing. This was also the case for the observations of Duysens and colleagues (1996). These earlier observations describe the current data fairly well. Here, reflex amplitudes and timings were quite similar, but there are some minor differences. For example the pattern shown for arm cycling seems close to a simple reversal that is much "cleaner" than that seen in leg muscles during walking. Perhaps this reflects the clear symmetry between directions in the arm cycling task (e.g., continuous task evenly distributed) which is not present during walking (e.g., continuous task with uneven distribution of activation for swing and stance).

It is notable that reflex amplitudes were uncoupled from background EMG irrespective of arm cycling direction. Previously it has been suggested that this is an indicator of the activity of CPG mechanisms during rhythmic movement (Haridas and Zehr 2003; Komiyama et al. 2000; Van Wezel et al. 1997; Zehr and Haridas 2003; Zehr and Kido 2001). The current data support the concept that the arms and legs are regulated by the same mechanisms during rhythmic motion that is independent of movement direction and that a portion of this control can be ascribed to CPG-like activity. Interestingly, data on human infants suggest a common CPG for walking that is independent of movement direction (Lamb and Yang 2000; Pang and Yang 2002). The current results thus extend to the human upper limb many of the features of reflex modulation observed during lower limb locomotor control and add further to the evidence suggesting similar control mechanisms (e.g., CPG contributions) to the generation of the rhythmic arm muscle activation pattern during arm cycling and walking.

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REFERENCES

Buford JA and Smith JL. Adaptive control for backward quadrupedal walking. II. Hindlimb muscle synergies. *J Neurophysiol* 64: 756–766, 1990.

Buford JA and Smith JL. Adaptive control for backward quadrupedal walking. III. Stumbling corrective reactions and cutaneous reflex sensitivity. *J Neurophysiol* 70: 1102–1114, 1993.

Burke RE. The use of state-dependent modulation of spinal reflexes as a tool to investigate the organization of spinal interneurons. *Exp Brain Res* 128: 263–277, 1999.

Burke RE, Degtyarenko AM, and Simon ES Patterns of locomotor drive to motoneurons and last-order interneurons: clues to the structure of the CPG. *J Neurophysiol* 86: 447–462, 2001.

Dietz V. Do human bipeds use quadrupedal coordination? *Trends Neurosci* 25: 462–467, 2002a.

Dietz V. Proprioception and locomotor disorders. *Nat Rev Neurosci* 3: 781–790, 2002b.

Dietz V, Fouad K, and Bastiaanse CM. Neuronal coordination of arm and leg movements during human locomotion. *Eur J Neurosci* 14: 1906–1914, 2001.

Duysens J and Tax T Interlimb reflexes during gait in cat and human. In: *Interlimb Coordination: Neural, Dynamical, and Cognitive Constraints,* edited by Swinnen SP, Heuer H, Massion J, and Casaer P. New York: Academic, 1994, p. 97–126.

Duysens J, Tax AA, Murrer L, and Dietz V. Backward and forward walking use different patterns of phase-dependent modulation of cutaneous reflexes in humans. *J Neurophysiol* 76: 301–310, 1996.

Duysens J and Van de Crommert HW. Neural control of locomotion. I. The central pattern generator from cats to humans. *Gait Posture* 7: 131–141, 1998.

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Eisner WD, Bode SD, Nyland J, and Caborn DN. Electromyographic timing analysis of forward and backward cycling. *Med Sci Sports Exerc* 31: 449–455, 1999.

Grasso R, Bianchi L, and Lacquaniti F. Motor patterns for human gait: backward versus forward locomotion. *J Neurophysiol* 80: 1868–1885, 1998.

Grillner S Control of locomotion in bipeds, tetrapods, and fish. In: *Handbook of Physiology. The Nervous System. Motor Control.* Bethesda, MD: Am. Physiol. Soc., 1981, sect. 1, vol. II, p 1179–1236.

Haridas C and Zehr EP. Coordinated interlimb compensatory responses to electrical stimulation of cutaneous nerves in the hand and foot during walking. *J Neurophysiol* 90: 2850–2861, 2003.

Komiyama T, Zehr EP, and Stein RB. Absence of nerve-specificity in human cutaneous reflexes during standing. *Exp Brain Res* 133: 267–272, 2000.

Lamb T and Yang JF. Could different directions of infant stepping be controlled by the same locomotor central pattern generator? *J Neurophysiol* 83: 2814–2824, 2000.

MacKay-Lyons M. Central pattern generation of locomotion: a review of the evidence. *Phys Ther* 82: 69–83, 2002.

Pang MY and Yang JF. Sensory gating for the initiation of the swing phase in different directions of human infant stepping. *J Neurosci* 22: 5734–5740, 2002.

Pearson KG. Common principles of motor control in vertebrates and invertebrates. *Annu Rev Neurosci* 16: 265–297, 1993.

Perell KL, Gregor RJ, Buford JA, and Smith JL. Adaptive control for backward quadrupedal walking. IV. Hindlimb kinetics during stance and swing. *J Neurophysiol* 70: 2226–2240, 1993.

Thorstensson A. How is the normal locomotor program modified to produce backward walking? *Exp Brain Res* 61: 664–668, 1986.

Ting LH, Kautz SA, Brown DA, and Zajac FE. Phase reversal of biomechanical functions and muscle activity in backward pedaling. *J Neurophysiol* 81: 544–551, 1999.

Van Wezel BM, Ottenhoff FA, and Duysens J. Dynamic control of location-specific information in tactile cutaneous reflexes from the foot during human walking. *J Neurosci* 17: 3804–3814, 1997.

Winter DA, Pluck N, and Yang JF. Backward walking: a simple reversal of forward walking? *J Mot Behav* 21: 291–305, 1989.

Yamaguchi T. The central pattern generator for forelimb locomotion in the cat. *Prog Brain Res* 143: 115–122, 2004.

Zehr EP, Carroll TJ, Chua R, Collins DF, Frigon A, Haridas C, Hundza SR, and Thompson AK. Possible contributions of spinal CPG activity to rhythmic human arm movement. *Can J Physiol Pharm* 82: 556–568, 2004.

Zehr EP and Chua R. Modulation of human cutaneous reflexes during rhythmic cyclical arm movement. *Exp Brain Res* 135: 241–250, 2000.

Zehr EP, Collins DF, Frigon A, and Hoogenboom N. Neural control of rhythmic human arm movement: phase dependence and task modulation of Hoffmann reflexes in forearm muscles. *J Neurophysiol* 89: 12–21, 2003.

Zehr EP and Duysens J Regulation of arm and leg movement during human locomotion. *The Neuroscientist* 10: 347–361, 2004.

Zehr EP and Haridas C. Modulation of cutaneous reflexes in arm muscles during walking: further evidence of similar control mechanisms for rhythmic human arm and leg movements. *Exp Brain Res* 149: 260–266, 2003.

Zehr EP, Hesketh KL, and Chua R. Differential regulation of cutaneous and H-reflexes during leg cycling in humans. *J Neurophysiol* 85: 1178–1185, 2001.

Zehr EP and Kido A. Neural control of rhythmic, cyclical human arm movement: task dependency, nerve specificity and phase modulation of cutaneous reflexes. *J Physiol* 537: 1033–1045, 2001.

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